

## **Stage 6 Science- Revision**

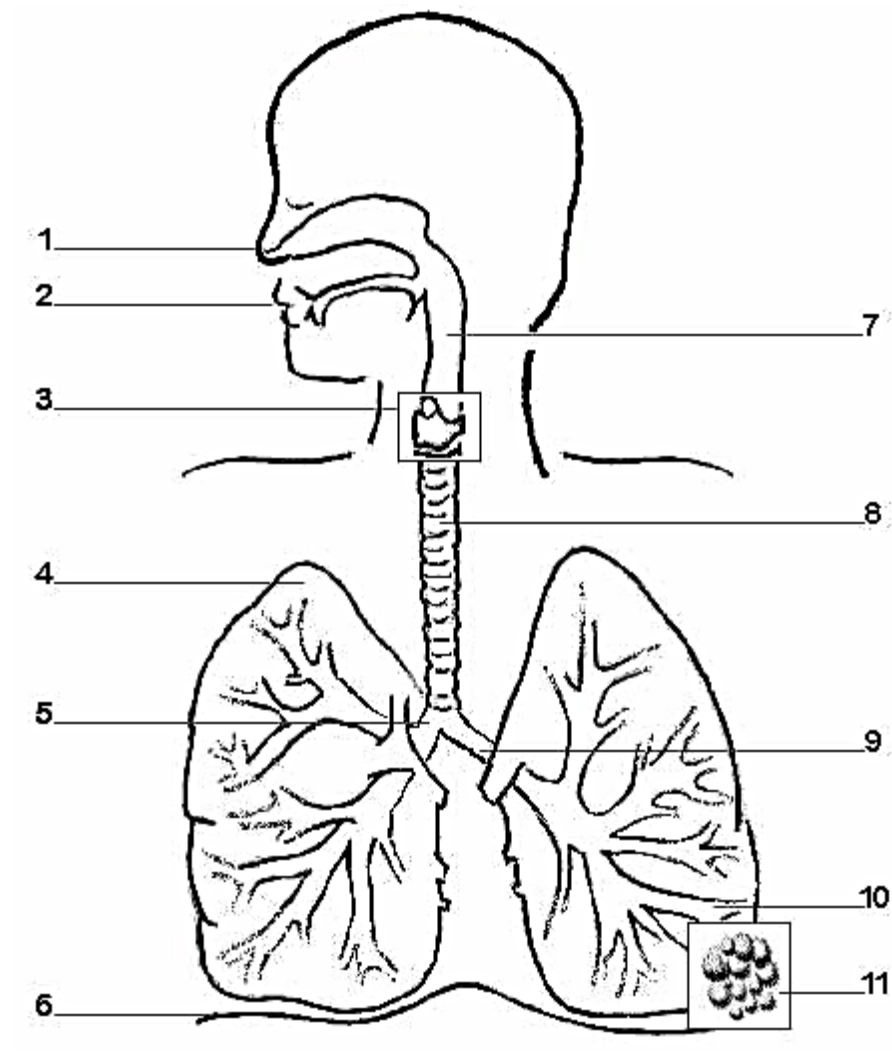
### **Respiratory System Vocabulary**

#### **Definitions**

- 1. respiration:** a process where air is taken in by the body and sent to the lungs, oxygen is extracted and sent to the rest of the body, energy is created by the cells using sugar and oxygen, carbon dioxide is released as a waste material and sent back to the lungs, and finally the carbon dioxide is released from the lungs
- 2. breathing:** the process of bringing in fresh air to the lungs and expelling "used" air
- 3. inhale:** when fresh air is brought into the lungs
- 4. exhale:** when "used" air is expelled from the lungs
- 5. intercostal muscles (rib muscles):** the muscles between the ribs that help lift the rib cage outward and upward during inhaling
- 6. diaphragm:** the large muscle at the base of the rib cage that contracts to make more room for the lungs during inhaling
- 7. respiratory tract:** the pathway that air takes while breathing
- 8. mouth/nose:** the main entry and exit points for air during breathing
- 9. nasal cavity:** the space behind the nose where air first enters from the outside of the nose
- 10. mucus:** the sticky, moist substance that traps some of the dirt and dust in the airways
- 11. pharynx (throat):** the area below the nasal cavities and behind the mouth where air travels before entering the larynx
- 12. epiglottis:** the flap that covers the larynx while swallowing food
- 13. larynx (voice box):** the area of the throat (pharynx) where sounds are produced for communication
- 14. trachea (wind pipe):** the main air tube in the respiratory tract that is located directly below the larynx
- 15. thoracic cavity (chest cavity):** the space inside your body which contains your heart and lungs and other related organs
- 16. bronchus:** one of the larger air tubes found in the airway after passing through the trachea

Read and answer the following

|                           |                      |                                                                                                                   |
|---------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Diaphragm</b>          | <input type="text"/> | Air sacs surrounded by capillaries where gaseous exchange takes place                                             |
| <b>Alveoli</b>            | <input type="text"/> | Attaches to the trachea and provides a pathway into the left and right lungs                                      |
| <b>Pharynx</b>            | <input type="text"/> | The smallest of the airways that are found inside the lungs and supply air to the alveoli                         |
| <b>Lungs</b>              | <input type="text"/> | Hang suspended behind the chest straddling the heart. This body part helps breathing by expanding and contracting |
| <b>Nasal Cavity</b>       | <input type="text"/> | Also known as the throat which provides a pathway for air into the tracheas                                       |
| <b>Larynx</b>             | <input type="text"/> | Space behind the nose which allows an entrance for air into the respiratory system                                |
| <b>Epiglottis</b>         | <input type="text"/> | A key muscle underneath the lungs which controls breathing                                                        |
| <b>Bronchioles</b>        | <input type="text"/> | Muscles attached to the ribs which help to move the rib cage during respiration                                   |
| <b>Trachea</b>            | <input type="text"/> | Contains the vocal cords which are surrounded by tough muscle fibres for protection                               |
| <b>Bronchus (Bronchi)</b> | <input type="text"/> | The windpipe which attaches to the Bronchi and is protected by tough cartilage rings                              |
| <b>Intercostals</b>       | <input type="text"/> | A flap of tissue that prevents food from entering into the trachea                                                |



Label the diagram

2. Write down the function of 1, 2 and 6?

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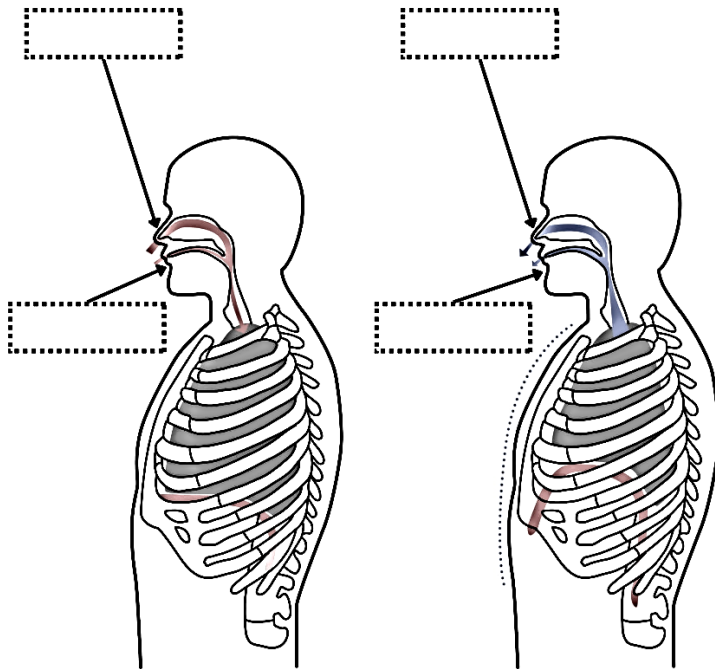
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Label the diagram, and explain the 2 (two) processes shown the diagram

## **Airflow through the respiratory system.**



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## **Title: Lungs**

### **Key question:**

What are the functions and structure of lungs?

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### **Activity: Making a lung model Results:**

1. What happens to the balloon when we pull the center of the rubber out?

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2. What happens to the balloon when we push the center of the rubber in?

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### **Discussion**

3: Which part of the lung model represents the lungs or the thin sheet of muscle?

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4. What is the work of the sheet of muscle?

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5. How does the sheet of muscle move?

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6.: Can you guess what happens to the sheet of muscle when we breath in or out?

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## Summary

- Lung is a respiratory organ, situated inside the rib cage, that transfer oxygen into the blood and remove carbon dioxide from it.
- Diaphragm is a special muscle that helps our lungs to move.
- When inhaling, the diaphragm moves down. This causes lungs to become bigger and allows air to come into lungs.
- When exhaling, the diaphragm moves up. This causes lungs to become smaller and air is forced out of the lungs.

The act of breathing has two stages – inhalation and exhalation

- Inhalation – the intake of air into the lungs through expansion of chest volume.
- Exhalation – the expulsion of air from the lungs through contraction of chest volume.
- Inhalation and exhalation involve a /muscle, which is called diaphragm muscle

### 1. During inhalation – the muscles contract:

- Contraction of the diaphragm muscle – causes the diaphragm to flatten, thus enlarging the chest cavity. **The chest cavity and lungs expand**, thus reducing air pressure and causing air to be drawn into the lungs.

### 2. During exhalation – the muscles relax:

- The muscles are no longer contracting, they are relaxed.
- The diaphragm curves and rises, the ribs descend and chest volume decreases.



Topic-



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What I know

What I want to know

What I have learnt

